Kernix 0.11.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

35 workloads

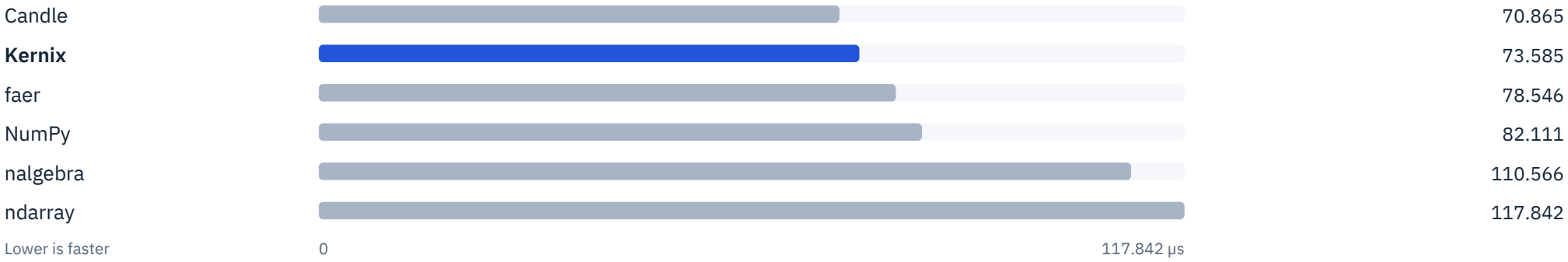
119 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



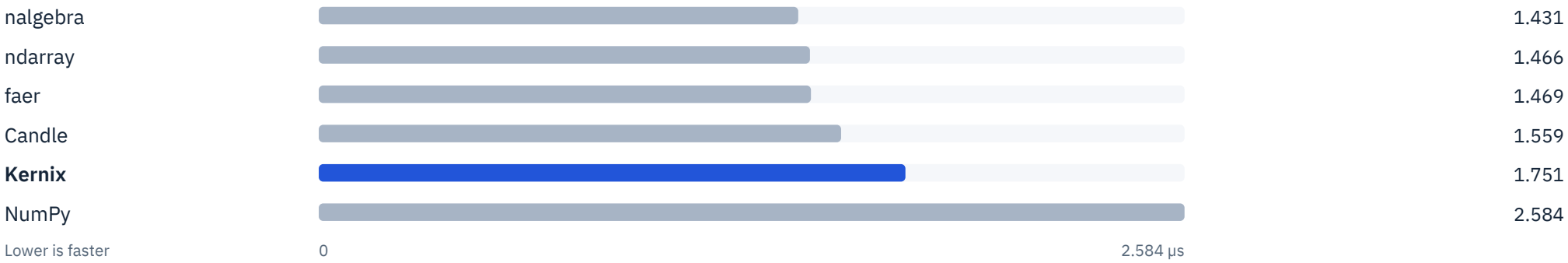
Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

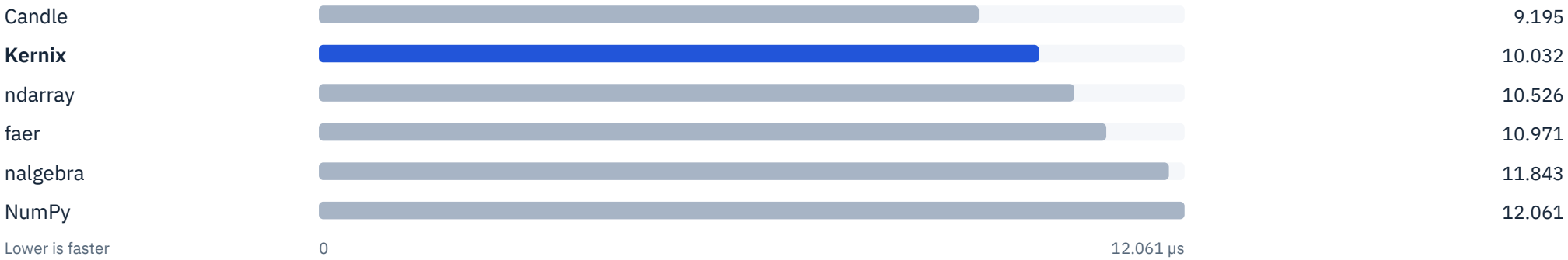
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



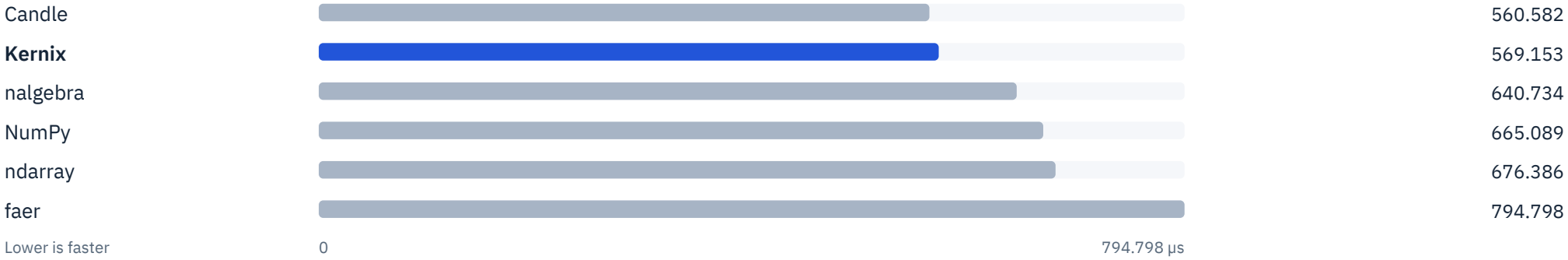
Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

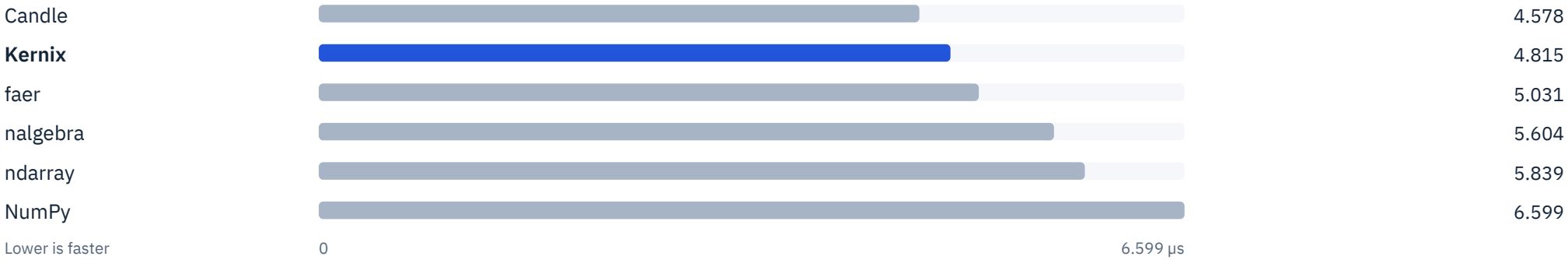
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

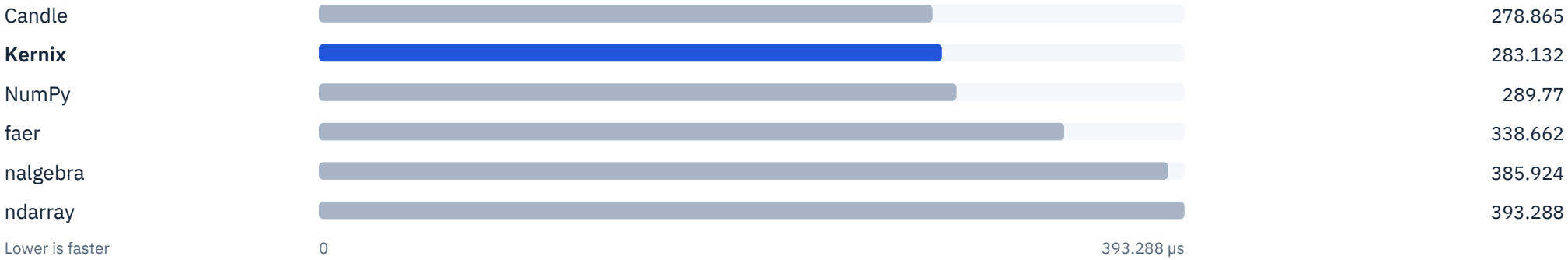
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



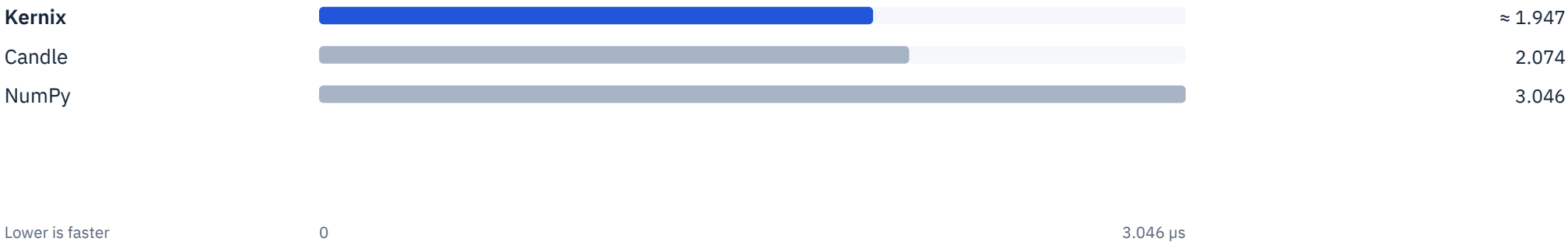
Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

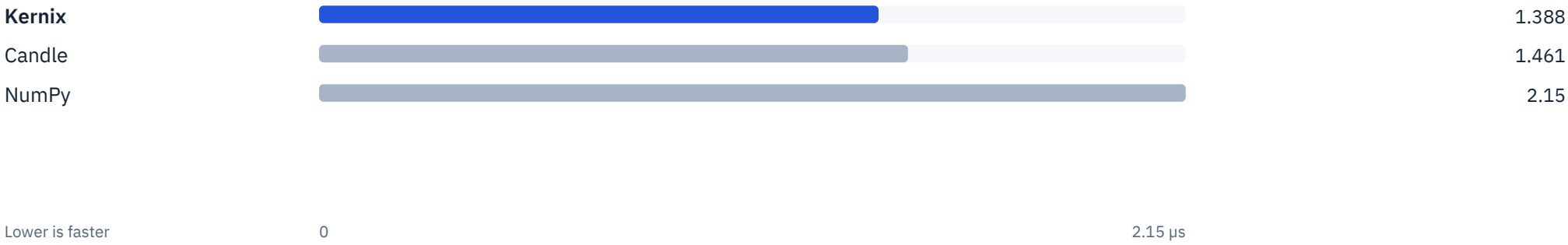
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

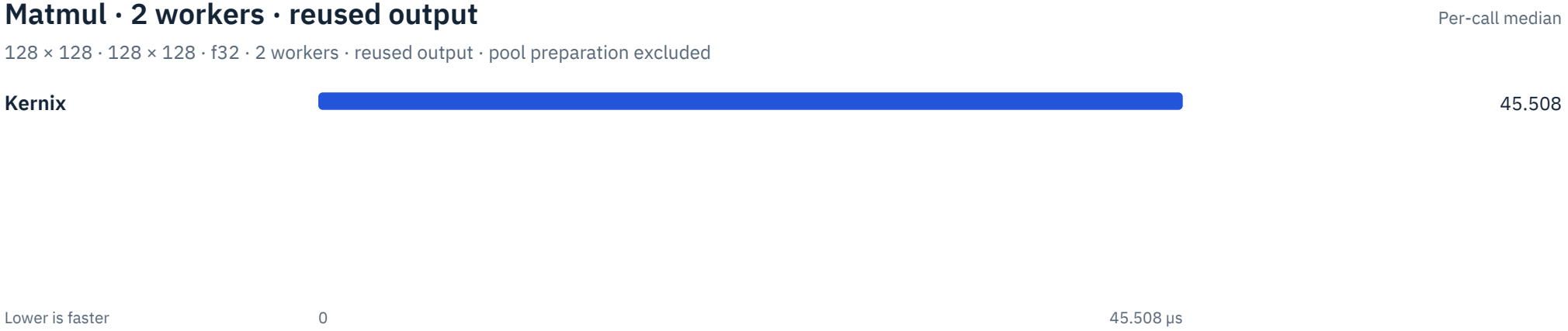
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



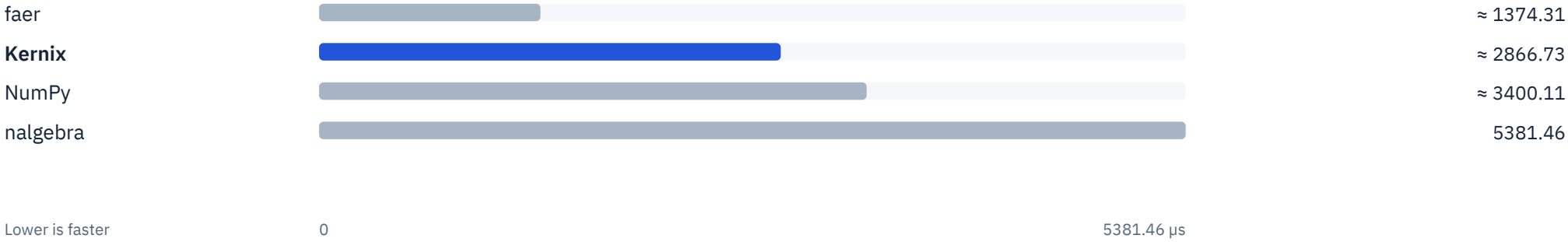
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

512 × 512 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



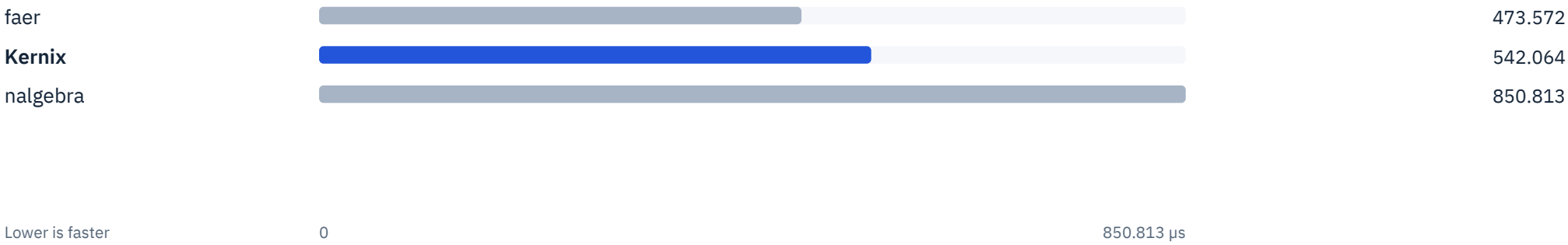
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

Per-call median

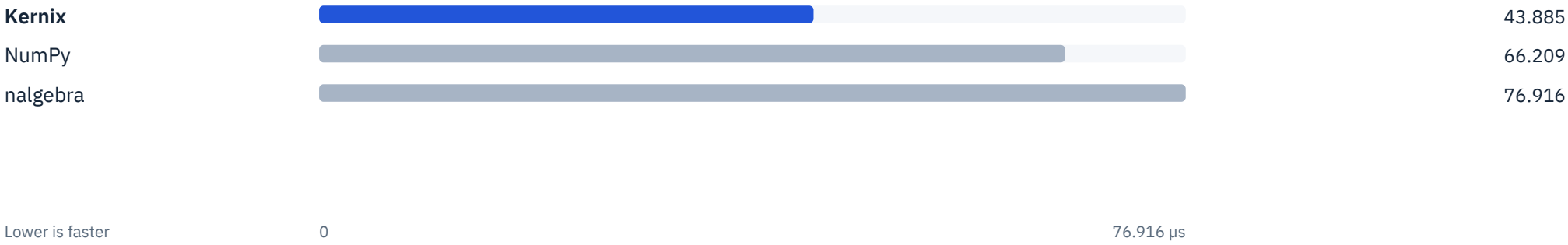
256 × 64 · f64 · 1 worker · allocated output



Minimum-norm least squares

Per-call median

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



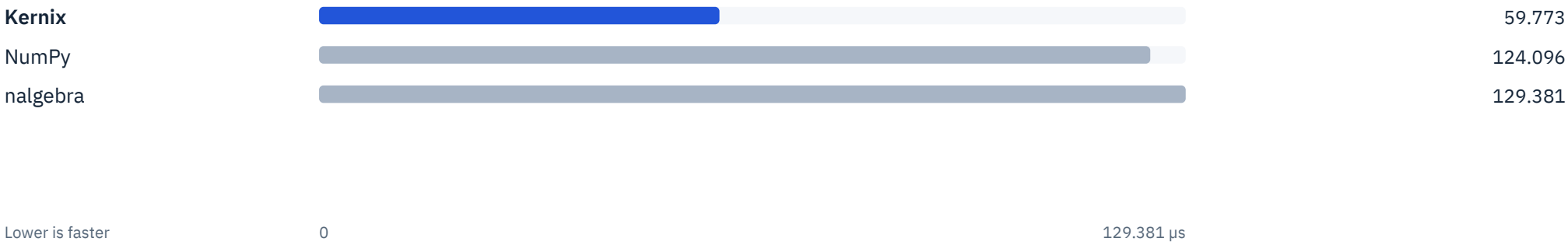
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

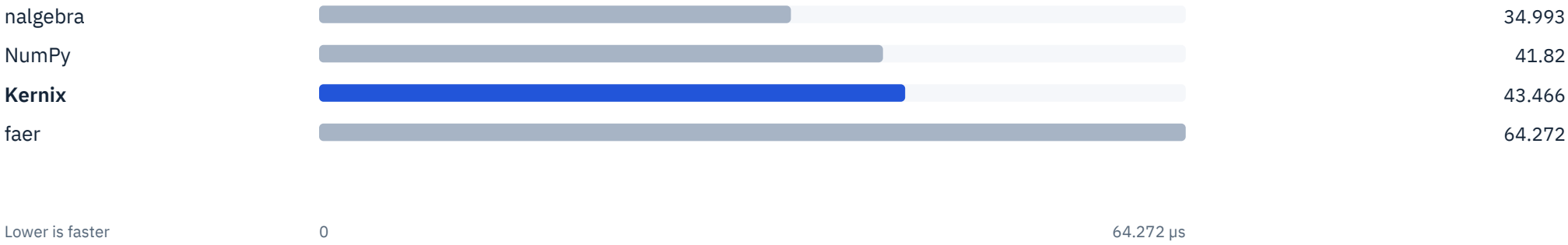
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

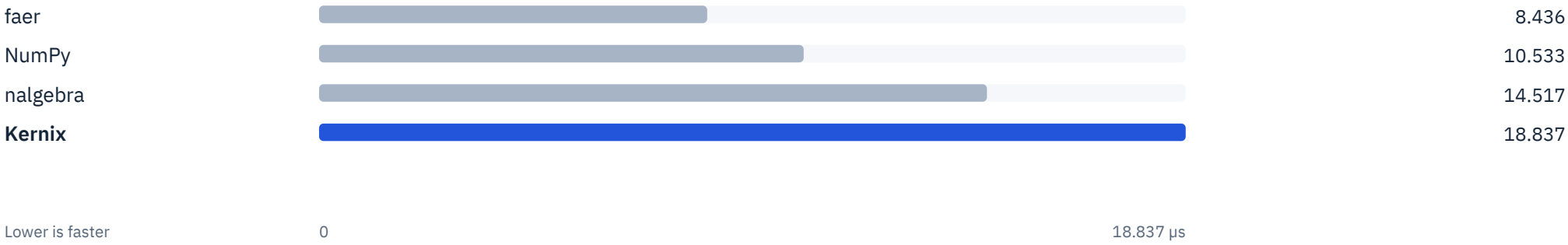
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



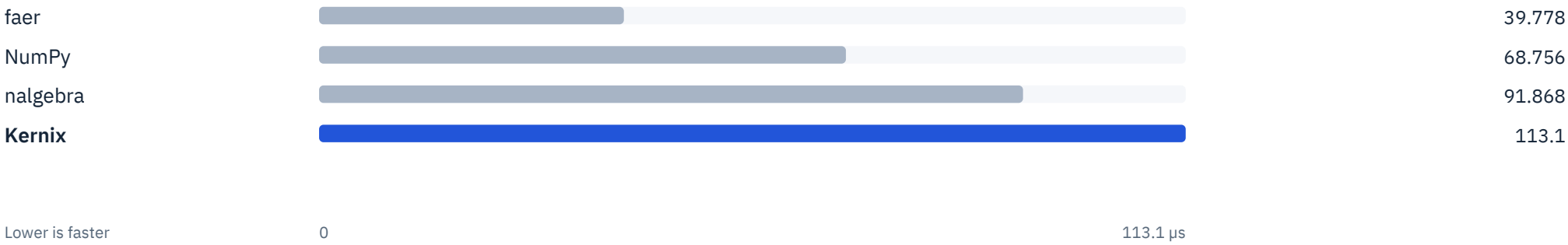
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

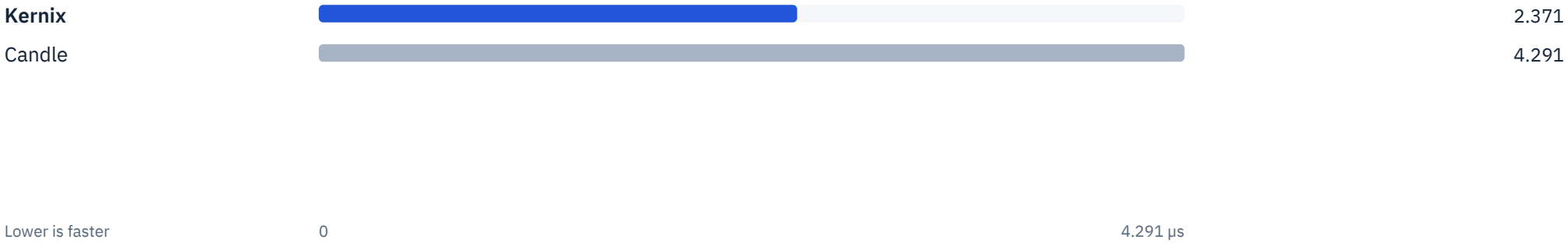
sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



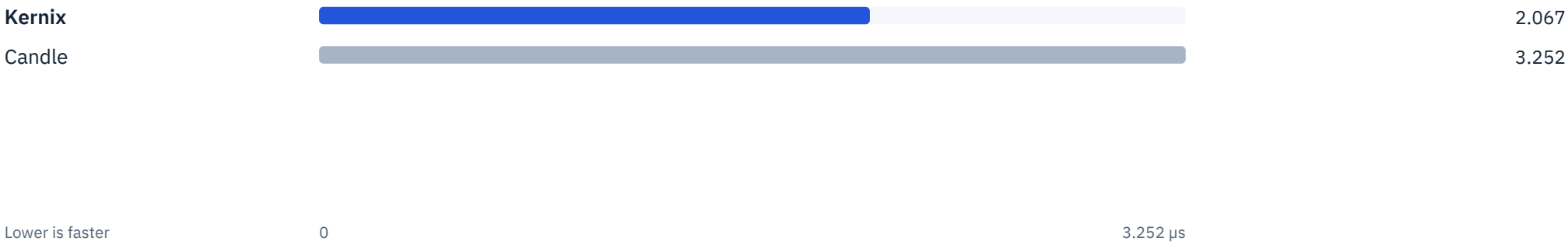
Kernix 0.11.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

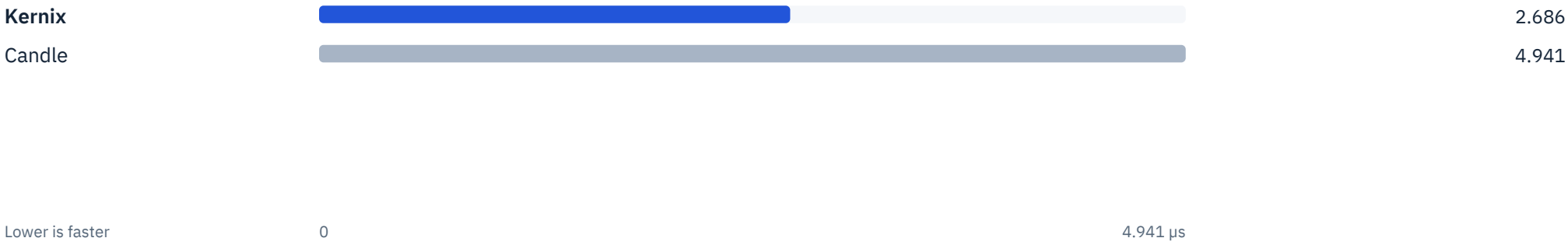
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



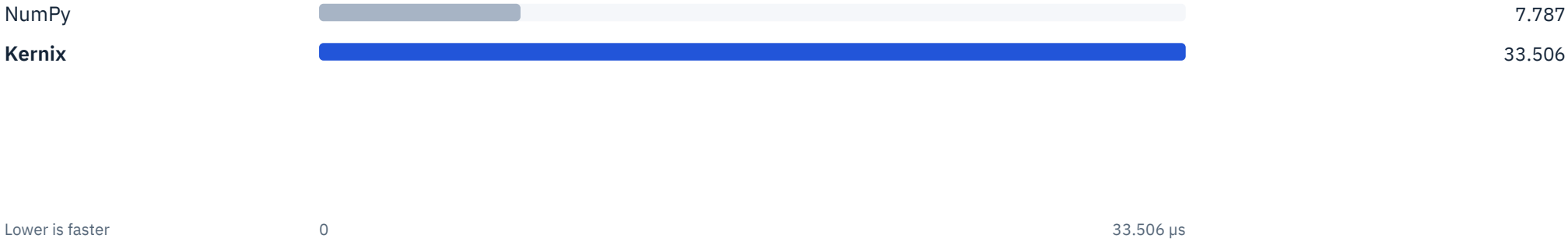
Kernix 0.11.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



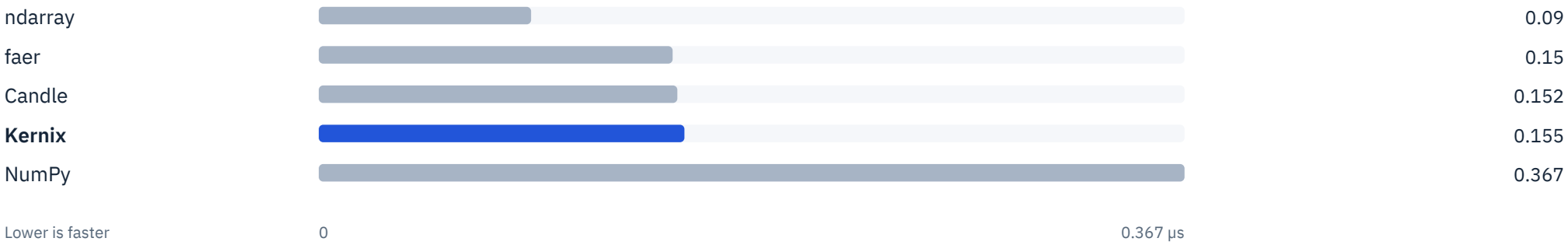
Kernix 0.11.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	73091	73.091
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	70656	70.656
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	37337	37.337
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	36784	36.784
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1559	1.559
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1466	1.466
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1431	1.431
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1469	1.469
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2584	2.584
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1751	1.751

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10526	10.526
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12061	12.061
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9195	9.195
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10971	10.971
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11843	11.843
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	10032	10.032
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	82111	82.111
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	117842	117.842
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	73585	73.585
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	78546	78.546

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	110566	110.566
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	70865	70.865
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	665089	665.089
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	560582	560.582
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	640734	640.734
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	676386	676.386
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	794798	794.798
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	569153	569.153
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	5839	5.839
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6599	6.599

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5031	5.031
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4815	4.815
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4578	4.578
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	5604	5.604
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	51639	51.639
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	41242	41.242
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	36027	36.027
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	48244	48.244
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	39073	39.073
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	36543	36.543

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	385924	385.924
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	289770	289.77
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	393288	393.288
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	338662	338.662
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	283132	283.132
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	278865	278.865
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2074	2.074
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3046	3.046
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1947.0000000000002	1.9470000000000002
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1461	1.461

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2150	2.15
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1388	1.388
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	9793	9.793
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	42230	42.23
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	295463	295.463
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	45508	45.508
Cholesky factorization / nalgebra 512 × 512 · f64	5381460	5381.46
Cholesky factorization / NumPy 512 × 512 · f64	3400107	3400.107
Cholesky factorization / faer 512 × 512 · f64	1374314	1374.314
Cholesky factorization / Kernix 512 × 512 · f64	2866731	2866.731

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	70295	70.295
Column-pivoted QR / nalgebra 96 × 32 · f64	76127	76.127
Column-pivoted QR / Kernix 96 × 32 · f64	58542	58.542
Column-pivoted QR / nalgebra 256 × 64 · f64	850813	850.813
Column-pivoted QR / faer 256 × 64 · f64	473572	473.572
Column-pivoted QR / Kernix 256 × 64 · f64	542064	542.064
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	76916	76.916
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	66209	66.209
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	43885	43.885
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	124096	124.096

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	59773	59.773
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	129381	129.381
Thin QR / faer 96 × 32 · f64	64272	64.272
Thin QR / NumPy 96 × 32 · f64	41820	41.82
Thin QR / Kernix 96 × 32 · f64	43466	43.466
Thin QR / nalgebra 96 × 32 · f64	34993	34.993
Thin QR / faer 256 × 64 · f64	420045	420.045
Thin QR / nalgebra 256 × 64 · f64	362227	362.227
Thin QR / Kernix 256 × 64 · f64	472111	472.111
Thin QR / NumPy 256 × 64 · f64	368509	368.509

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	18837	18.837
Cholesky factorization / faer 64 × 64 · f64	8436	8.436
Cholesky factorization / NumPy 64 × 64 · f64	10533	10.533
Cholesky factorization / nalgebra 64 × 64 · f64	14517	14.517
Cholesky factorization / NumPy 128 × 128 · f64	68756	68.756
Cholesky factorization / nalgebra 128 × 128 · f64	91868	91.868
Cholesky factorization / faer 128 × 128 · f64	39778	39.778
Cholesky factorization / Kernix 128 × 128 · f64	113100	113.1
Cholesky factorization / NumPy 256 × 256 · f64	511409	511.409
Cholesky factorization / faer 256 × 256 · f64	210253	210.253

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	695933	695.933
Cholesky factorization / Kernix 256 × 256 · f64	503917	503.917
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3885.9999999999995	3.8859999999999995
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6122	6.122
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4291	4.291
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2371	2.371
Model inference / Candle batched normalized/sliced MLP · f64	3252	3.252
Model inference / Kernix batched normalized/sliced MLP · f64	2067	2.067
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2686	2.686
Model recording + forward / Candle batched normalized/sliced MLP · f64	4941	4.941

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	3032	3.032
Model backward / Candle prepared graph → fresh gradients · f64	14518	14.518
Parameter update / Kernix transactional SGD, fixed gradients · f64	560	0.56
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	33506	33.506
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	7787	7.787
Row softmax / Candle 256 × 64 · f64	77863	77.863
Row softmax / ndarray 256 × 64 · f64	70836	70.836
Row softmax / Kernix 256 × 64 · f64	85747	85.747
Row softmax / NumPy 256 × 64 · f64	96953	96.953
Create zeros / faer 64 × 64 · f32	150	0.15

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / NumPy 64 × 64 · f32	367	0.367
Create zeros / Kernix 64 × 64 · f32	155	0.155
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create ones / Candle 64 × 64 · f32	243	0.243
Create ones / ndarray 64 × 64 · f32	140	0.14
Create ones / NumPy 64 × 64 · f32	1278	1.278
Create ones / Kernix 64 × 64 · f32	184	0.184
Create ones / faer 64 × 64 · f32	188	0.188